

A Quantal Response Analysis of Simultaneous Multi-Target Attacker-Defender Security Games

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Abstract—Modern systems, under the management of human decision-makers, confront rapidly increasing cybersecurity threats. This paper considers security decision-making within these complex modern systems managed by human defender. The defender has responsibility for safeguarding a specific set of assets with heterogeneous loss valuations against an external attacker. These assets are associated with probabilities that represent the likelihood of a successful attack, which can be reduced through security investments by the defender and can be increased via attacker's investments. Our approach involves modeling these systems using simultaneous game-theoretic framework, accounting for the impact of bounded rationality and imperfect best-response behavior—as frequently observed in human decision-making within the domains of behavioral economics and psychology. We first establish the existence of a quantal response equilibrium in our security games. Subsequently, we analyze the effects of this type of bounded rationality in terms of the expected cost of the defender. To assess our models, we employ a representative system and compare the game-theoretic optimal investment strategies under those derived from a quantal response equilibrium standpoint. We also investigate the influence of various system parameters on the overall security level (expected security cost) of this system.

Index Terms—Quantal response equilibrium, CyberSecurity, Human decision-making, Heterogeneous assets, Game Theory.

I. INTRODUCTION

Modern systems face an ever-increasing threat of sophisticated cyberattacks performed by external adversaries. These attacks pose significant risks to critical infrastructures on a large scale [1]. The primary objective behind such attacks is to gain control of vital assets within these systems [2]. To achieve this goal, attackers often exploit various vulnerabilities associated with each critical asset [3]. In this complex landscape, security decision-makers responsible for managing these systems must make wise allocations of their limited security budgets to mitigate security risks [4].

Although there is a massive body of security literature capturing security resource allocation problems in modern systems, most of the existing work has relied on *classical models* of human decision-making (with few exceptions that explored the impact of human prospect-theoretic attitudes for security games [4]–[7]), where all defenders and attackers are assumed to make fully rational risk evaluations, perceptions, and security decisions [8]. On the contrary, behavioral economics and psychology have shown empirically that

humans consistently deviate from these classical (fully rational) models of decision-making.

Notably, research in *behavioral economics and psychology* has shown that humans consistently make errors in choosing efficient (pure) strategies when making decisions [9]. In contrast to classical decision-making, the quantal response equilibrium takes into account the fact that human decision-makers may not always make perfectly rational decisions. Instead, it allows for decision-making where the players in a game may not always choose the best response with certainty, but their choices are probabilistic and influenced by factors such as noise and cognitive limitations. Many empirical studies have provided evidence for this class of behavioral models (e.g., [10], [11]). The effects of this quantal behavioral decision-making are relevant for evaluating security of modern systems in which security decisions are made through human decision-making, albeit with help from threat assessment tools [12], [13].

The majority of prior works in security resource allocation for modern systems have considered the Pure-strategy Nash equilibrium (PNE) solution concept in which the search strategy for the best resource allocation for each decision-maker adopts best response dynamics [7], [14]. However, behavioral economics and psychology have shown that humans have errors in choosing which pure strategy to select in real-world scenarios [10], [11], [15]. Such a process can be modeled using quantal functions (such as the logit function [11]) where the human chooses each strategy with a probability that is positively related to the payoff from that strategy. This process typically leads to quantal response equilibrium. There are several prior works that have shown such quantal response equilibrium for security problems, however with focus on Stackelberg security games [16], and attacker with bounded rationality behavior [17].

In contrast to such studies, we consider general defense allocation schemes that can be applied to any system where failure scenarios are represented by multiple targets with heterogeneous loss valuations while modeling human behavioral decision-making. Our work introduces a rigorous framework that combines two important threads of human decision-making for systems security research: game theory (modeled by attacker-defender simultaneous security games on

multiple assets with heterogeneous valuations), and behavioral economics (modeled by quantal response equilibrium).

We first model the effect that behavioral decision-making (quantal response) has on the simultaneous attacker-defender game on systems with multiple assets. In these systems, the defender is responsible for defending a set of critical assets against external adversary [18]. We formulate such a scenario as a *behavioral multi-target security game*. We show that such a game has a quantal response equilibrium (QRE). We then calculate the QRE where the search strategy for each decision-maker adopts quantal response dynamics [10], [11]. We finally evaluate our findings using a representative system that has four assets (targets) with heterogeneous loss valuations. In our evaluation, we show the effects of different parameters (e.g., financial loss values of system's assets, and behavioral level of players).

In summary, this paper makes the following contributions:

- 1) We propose a *security investment* model for modern systems where human defender's assets have heterogeneous loss valuations and interaction with attacker is captured by simultaneous security game.
- 2) We show the effect of an important human behavioral error (probabilistic and noisy best response as modeled as quantal response equilibrium) on security resource allocation decision-making of human defender for securing modern systems.
- 3) We illustrate the effects of *behavioral* errors of human decision-making on system security through a representative system. We also analyze different system parameters under our behavioral model.

II. THE MULTI-TARGET SECURITY GAME FRAMEWORK

In this section, we present our simultaneous multi-target security games, establishing a theoretical basis that can be used to model any two-player simultaneous move game formulation between the defender $\mathcal{D}$, and the attacker $\mathcal{A}$. A simple example of our setup is shown in Figure 1, which illustrates a system with a defender and an external attacker who seeks to exploit vulnerabilities within the network in order to reach and compromise critical targets [13]. Here, We introduce a defender and an attacker model, the defender's cost and attacker's utility, respectively.

A. Strategic Defender

Consider a defender $\mathcal{D}$ who is tasked with protecting a set of assets $V = \{v_1, v_2, \dots, v_n\}$. If an asset $v_m \in V$ is compromised, the defender will suffer a financial loss of $L_m \in \mathbb{R}_{>0}$. To lower the probabilities that the assets are successfully attacked, the defender can allocate her limited security budget to these assets. We assume that defender $\mathcal{D}$ has a security budget $B \in \mathbb{R}_{>0}$, which can be money, resources or a combination of both. Hence, the defense strategy space of the defender is defined as

$$X \triangleq \{\mathbf{x} \in \mathbb{R}_{\geq 0}^n : \sum_{v_i \in V} x_i \leq B\}. \quad (1)$$

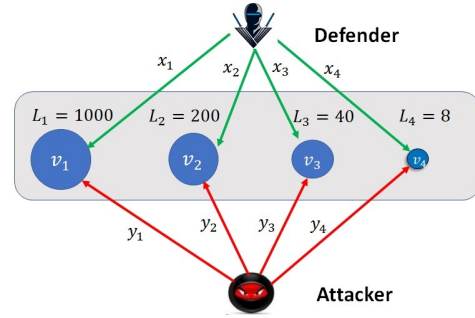


Fig. 1: An illustration of our Simultaneous Multi-Target Attacker-Defender game. The defender investments and the attacker investments are indicated respectively by green and red arrows on assets. x_i and y_i denote the amount of resources invested in defending and attacking asset v_i , respectively.

In words, the defense strategy space for defender $\mathcal{D}$ consists of all non-negative investments on assets such that the sum of all investments does not exceed the budget B . We denote any particular vector of investments by defender $\mathcal{D}$ by $\mathbf{x} \in X$.

B. Strategic Attacker

Consider an attacker $\mathcal{A}$ trying to compromise the assets set V . For each compromised asset $v_m \in V$, the attacker $\mathcal{A}$ will achieve a financial gain $G_m \in \mathbb{R}_{>0}$. To increase the attack success probabilities on assets, the attacker can allocate attack investments on these assets, subject to a budget constraint $P \in \mathbb{R}_{\geq 0}$. We define the attack strategy space of the attacker by

$$Y \triangleq \{\mathbf{y} \in \mathbb{R}_{\geq 0}^n : \sum_{v_i \in V} y_i \leq P\}. \quad (2)$$

We denote the attacker's investment vector by $\mathbf{y} \in Y$.

C. Defender's Cost and Attacker's Utility

We suppose that the probability of successful attack on an asset $v_i \in V$ is given by $p_i(x_i, y_i) : \mathbb{R}_{\geq 0}^2 \rightarrow [0, 1]$. Thus, the defender's investment x_i and the attacker's investment y_i on each asset v_i would affect the probability $p_i(x_i, y_i)$ that the asset can be successfully compromised by the attacker.

Defender's Cost: The defender $\mathcal{D}$ aims to choose defense investment vector $\mathbf{x}$ to best protect the assets from attacks. This goal is represented mathematically by the cost function

$$C_{\mathcal{D}}(\mathbf{x}, \mathbf{y}) = \sum_{v_i \in V} L_i p_i(x_i, y_i) \quad (3)$$

subject to $\mathbf{x} \in X$. For any given attacker investment $\mathbf{y} \in Y$, the defender chooses $\mathbf{x} \in X$ to minimize $C_{\mathcal{D}}(\mathbf{x}, \mathbf{y})$.

Attacker's Utility: The attacker $\mathcal{A}$ wants to choose attack investment vector $\mathbf{y}$ to successfully compromise assets. This is mathematically represented by the utility function

$$U_{\mathcal{A}}(\mathbf{x}, \mathbf{y}) = \sum_{v_i \in V} G_i p_i(x_i, y_i) \quad (4)$$

subject to $\mathbf{y} \in Y$. For any given defense investment $\mathbf{x} \in X$, the attacker chooses $\mathbf{y} \in Y$ to maximize $U_{\mathcal{A}}(\mathbf{x}, \mathbf{y})$.

Note that $C_{\mathcal{D}}(\mathbf{x}, \mathbf{y})$ and $U_{\mathcal{A}}(\mathbf{x}, \mathbf{y})$ depend on both the defense budget allocation $\mathbf{x}$ and the attack investment $\mathbf{y}$.

Figure 1 shows a simple example of our Simultaneous Multi-Target Security Game Setup.

Pure-strategy Nash Equilibrium: In a Multi-Target Simultaneous Security Game, a collection of best response strategies $(\mathbf{x}^*, \mathbf{y}^*)$ is a Pure-strategy Nash Equilibrium (PNE) if and only if both of equations (5) and (6) below are satisfied simultaneously:

$$\mathbf{x}^* \in \underset{\mathbf{x} \in X}{\operatorname{argmin}} C_{\mathcal{D}}(\mathbf{x}, \mathbf{y}^*) \quad (5)$$

$$\mathbf{y}^* \in \underset{\mathbf{y} \in Y}{\operatorname{argmax}} U_{\mathcal{A}}(\mathbf{x}^*, \mathbf{y}). \quad (6)$$

D. Probability of Successful Attack

A particular success function which we will focus on throughout this work is

$$p_i(x_i, y_i) = \exp(-x_i - a_i)(1 - \exp(-y_i)) \quad (7)$$

where $a_i \in \mathbb{R}_{\geq 0}$ in (7) represents the pre-existing (or inherent) security investments on the node v_i , which decrease the successful attack probability even under no additional defense investments. This probability function falls within the class commonly considered in security economics [19], [20].

III. QUANTAL RESPONSE EQUILIBRIUM

A. Quantal Response Modeling

Quantal response equilibrium (QRE) is a solution concept in game theory which provides an equilibrium concept with bounded rationality [21]. It provides a structural, statistical model of human operators where humans consistently make errors in choosing efficient strategies as shown in behavioral economics and psychology when modeling human decision-making (e.g., [10], [11]). In contrast to the deterministic and perfect best-response behavior of PNE, in the QRE players “better respond” and choose strategies that provide higher payoffs with a higher probability.

In our simultaneous defender-attacker security game logit equilibrium, the defender’s and attacker’s security investment profiles would be chosen according to the following probability distributions

$$\sigma_{dl} = \frac{\exp(-\lambda_d EC_{dl}(\sigma_a))}{\sum_{\mathbf{x}_l \in X} \exp(-\lambda_d EC_{dl}(\sigma_a))}, \quad (8)$$

$$\sigma_{al} = \frac{\exp(\lambda_a EU_{al}(\sigma_d))}{\sum_{\mathbf{y}_l \in Y} \exp(\lambda_a EU_{al}(\sigma_d))}, \quad (9)$$

where σ_{dl} and σ_{al} are the probabilities of defender $\mathcal{D}$ choosing investment profile $\mathbf{x}_l \in X$ and attacker $\mathcal{A}$ choosing investment profile $\mathbf{y}_l \in Y$, respectively. Note that $EC_{dl}(\sigma_a)$ is the expected cost of defender $\mathcal{D}$ choosing investment strategy $\mathbf{x}_l$ under the belief that the attacker $\mathcal{A}$ is playing according to the probability distribution σ_a which is given by

$$EC_{dl}(\sigma_a) = \sum_{\mathbf{y}_j \in Y} \sigma_{aj} C_{\mathcal{D}}(\mathbf{x}_l, \mathbf{y}_j).$$

Similarly, $EU_{al}(\sigma_d)$ is the expected utility of attacker $\mathcal{A}$ of choosing investment strategy $\mathbf{y}_l$ under the belief that the

defender $\mathcal{D}$ is playing according to the probability distribution σ_d which is given by

$$EU_{al}(\sigma_d) = \sum_{\mathbf{x}_j \in X} \sigma_{dj} U_{\mathcal{A}}(\mathbf{x}_j, \mathbf{y}_l).$$

Here, the probability of defender $\mathcal{D}$ choosing $\mathbf{x}_l$ (σ_{dl}) increases if the expected cost for the defender decreases under that $\mathbf{x}_l$ and for the attacker $\mathcal{A}$ the probability (σ_{al}) increases with the increase of the expected gain for attacker choosing $\mathbf{y}_l$.

Based on such quantal responses, our search seeks to find fixed points to achieve quantal response equilibrium (σ_d^*, σ_a^*) as in mean field theory [10]. In our evaluation, we show the effects of the non-negative parameters λ_d and λ_a which represent the rationality level of defender $\mathcal{D}$ and attacker $\mathcal{A}$, respectively. When $\lambda_d, \lambda_a \rightarrow 0$, the players become “completely non-rational (behavioral)” and chooses each investment profile with equal probability. As $\lambda_d, \lambda_a \rightarrow \infty$, players become “perfectly rational” and the game approaches a Pure-strategy Nash equilibrium.

Remark 1. The subscripts d and a in λ_d and λ_a allow each player in our multi-target security game to have a different level of rationality. Throughout the rest of the paper, we will drop these subscripts when it is clear from the context. ■

B. Existence of a QRE

We first establish the existence of a QRE for the class of simultaneous attacker-defender game defined in Section III. Recall that a mixed strategy of profiles of security investments by the defenders is said to be a QRE when players assign probabilities to their available security investment profiles, representing the likelihood of selecting each profile. These probabilities are determined by the players’ quantal response functions (8) and (9), which describe how sensitive players are to the differences in expected costs and expected gain between different defense and attack investment strategies, respectively.

Proposition 1. Suppose the quantal response function for defender $\mathcal{D}$ is given by (8) and the quantal response function for attacker $\mathcal{A}$ is given by (9). Suppose that the attack success function $p_i(x_i, y_i)$ is given by (7) for every asset $v_i \in V$. Then, the Multi-target Security Game possesses a QRE when $\lambda_k \in [0, \infty)$ where $k \in \{d, a\}$.

Proof. The logit quantal response functions in (8) and (9) are interior, continuous, monotonic, and responsive [21]. Therefore, they are regular quantal response functions [21]. As a result, the multi-target Security Game considered in our work possesses a quantal response equilibrium (QRE). □

C. Mathematical Analysis Illustration of Finding QRE

Having provided our quantal response model and shown the existence of QRE for our simultaneous multi-target attacker-defender security game, we start by performing a mathematical quantal response analysis for a simplified example with a defender and an attacker with two strategies each to explain the concept of quantal response in our context.

TABLE I: Investment Strategies by defender and attacker.

(a) Investment strategies for defender $\mathcal{D}$

Strategy \ Asset Invest	v_1	v_2	v_3	v_4
X_1	9	1	0	0
X_2	8.2	1.8	0	0
X_3	7	2.8	0.2	0
X_4	5.9	3.3	0.8	0
X_5	4.9	3.4	1.6	0.1

(b) Investment strategies for Attacker $\mathcal{A}$

Strategy \ Asset Invest	v_1	v_2	v_3	v_4
Y_1	5	3.3	1.7	0
Y_2	3.6	3.9	2.5	0
Y_3	0.9	2.3	4.2	2.6
Y_4	1	2.2	4.2	2.6
Y_5	2.5	2.5	2.5	2.5

The goal of limiting the strategies in this mathematical analysis is to give the reader the main insight about how the QRE solution work when applied into our games. We refer to Section IV that shows our main mathematical results and experimental evaluation findings under larger set of investment strategies for both players (attacker and defender).

Consider the multi-target game setup of Figure 1, which has two players. Here, the defender is protecting the assets $v_i \in V$ and the attacker is trying to compromise the respective assets. We assume that defender $\mathcal{D}$ has investment strategies X_1 and X_5 , and attacker $\mathcal{A}$ has strategies Y_1 and Y_5 as shown in Table I. The probability of successfully compromising asset v_i is given by (7). To simply focus on the effect of quantal behavioral response, we consider $a_i = 1$ in this example. We first calculate the utilities (the cost for defender $\mathcal{D}$ and gain for attacker $\mathcal{A}$) under joint investment strategies for both players. Then, we calculate the quantal response equilibrium of the players' joint strategies under different behavioral levels (λ).

Consider the losses for defender and gains for attacker are given by the vector $[1000, 200, 40, 8]$ in the four assets of our setup (as shown in Figure 1). The four utilities of joint investment strategies by both players are calculated using (3) and (4) as follows:

	Y_1	Y_5
X_1	38.14, 38.14	41.10, 41.10
X_5	7.51, 7.51	9.94, 9.94

Let p be the probability that defender $\mathcal{D}$ chooses investment strategy X_5 and q be the probability that attacker $\mathcal{A}$ chooses attack strategy Y_5 . Then, the quantal response of the defender $\mathcal{D}$ to a given mixed strategy by attacker (parameterized by q) is to choose investment strategy X_5 with probability p and X_1 with $(1-p)$. In particular, when applying the quantal response function in (8), such probabilistic choice of strategy X_5 by the

defender $\mathcal{D}$ would be given by

$$p = \frac{\exp(-\lambda \cdot (9.94q + 7.51(1-q)))}{\exp(-\lambda \cdot (41.1q + 38.14(1-q))) + \exp(-\lambda \cdot (9.94q + 7.51(1-q)))}. \quad (10)$$

Similarly, the quantal response of the attacker $\mathcal{A}$ to given mixed strategy by defender (parameterized by p) would be to choose attack investment strategy Y_5 with probability q and Y_1 with $(1-q)$. Using the mixed investment strategy of attacker and the quantal response function in (9), we obtain the attacker's probabilistic choice of strategy Y_5 which is

$$q = \frac{\exp(\lambda \cdot (9.94p + 41.1(1-p)))}{\exp(\lambda \cdot (7.51p + 38.14(1-p))) + \exp(\lambda \cdot (9.94p + 41.1(1-p)))}. \quad (11)$$

Main Intuitions: Figure 2 shows the quantal response of the players under different behavioral levels. Such quantal response is given by the intersection of the two solid curves (red and blue lines), where each line shows the joint investment strategy for one of the players. These solid curves are obtained by plotting the equations (10) and (11). These figures show the following insights. First, Figure 2a shows that each player chooses randomly from the two investment strategies from their respective investment strategy spaces with equal probabilities when the players are highly behavioral (with very low λ , here $\lambda = 10$). Second, Figure 2b shows that as players become more rational (where λ increases), they choose better defense or attack strategies (i.e., those with lower cost for defender $\mathcal{D}$ and higher gain for attacker $\mathcal{A}$, respectively). Finally, Figure 2c emphasizes that QRE achieves PNE (from best responses) for the given strategies for extremely high values of λ . The evolution of the effect of behavioral level (λ) on the QRE solutions is evident in this analysis. The mathematical analysis and graphical representation of this example show that the players act randomly when they are more behavioral (with low λ) and they act rationally (according to best response or PNE solution) when they are more rational (with very high λ).

IV. EVALUATION

We now evaluate our multi-target simultaneous attacker-defender game. Our evaluation aims to answer the following research questions:

- What is the effect of quantal noise on the overall security level of the system (i.e., expected cost of defender)?
- How does the loss valuation of the critical assets affect the overall security level of the system with behavioral decision-making?
- What is the effect of simultaneous investment choices by defender and attacker on the system's security level (defender's expected cost) ?

A. Experimental Setup

We emulated a system with four critical assets (targets). For the defender, the first asset has very high loss (i.e., $L_1 = 1000$) while the second and third assets have lower losses

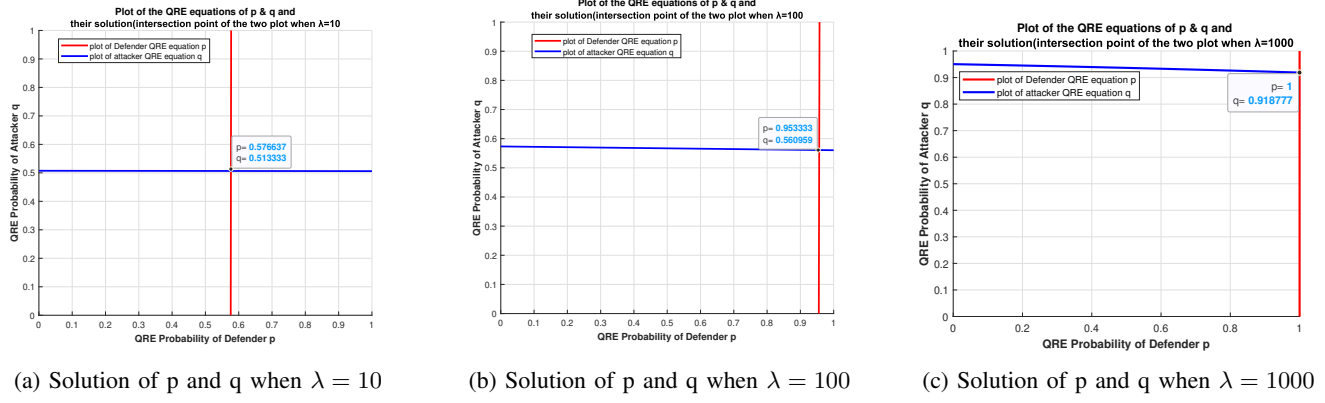


Fig. 2: Quantal response analysis of one defender and one attacker with two investment strategies each for our mathematical analysis illustration. In total, we show the effect of behavioral level on the probability of choosing security defense investments.

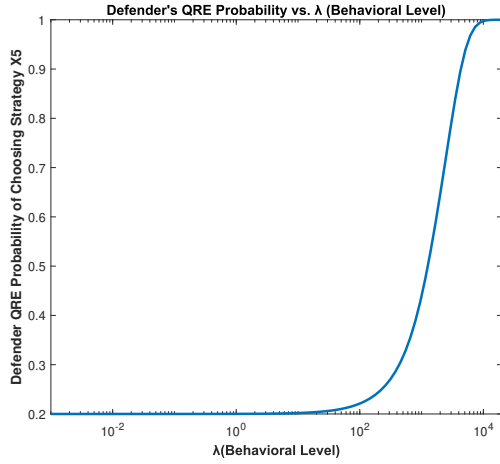


Fig. 3: QRE probability of Defender's investment strategy X_5 with varying behavioral levels.

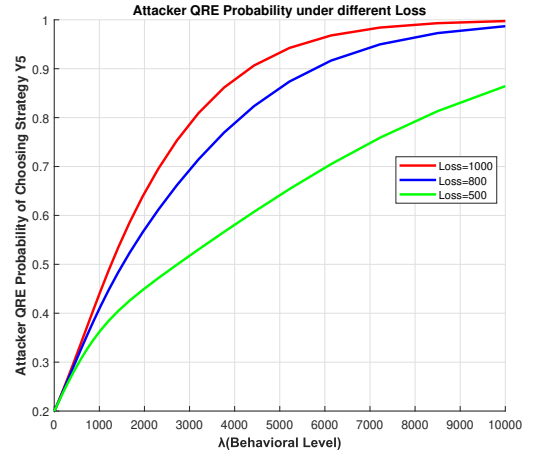


Fig. 4: QRE solution of attacker under different gains for successful compromise of the assets in the experimental setup.

(with $L_2 = 200, L_3 = 40$) and the fourth asset has the least loss ($L_4 = 8$). For the attacker, we employ symmetric gains for successful attack (i.e., $G_1 = 1000, G_2 = 200, G_3 = 40$, and $G_4 = 8$). We let the total defense budget for defending the four critical assets and the total attack budget to compromise them be $B = 10$ and $P = 10$, respectively. The probability of successful attack on each of the assets is given by

$$p(x, y) = e^{-x-1}(1 - e^{-y}) \quad (12)$$

where x and y are the defense and attack investment on that asset, respectively. We consider the pre-existing security investments on assets, $a_i = 1$. Table I shows the two investment strategy spaces (with five investment strategies each) for the defender $\mathcal{D}$ and the attacker $\mathcal{A}$. We use the best response dynamics notion to calculate these investments of each player at the PNE (very high λ). All of these investment strategies were calculated using Matlab Optimization toolbox.

B. Evaluation Results

We now summarize the main findings of our evaluation.

Effect of Behavioral Level (λ): Figure 3 illustrates this relationship of QRE probabilities with different players'

behavioral levels.¹ We observe that when increasing the behavioral level (λ) of the players, the QRE probabilities approach the best responses (i.e., which shows that players become more rational).

Effect of Financial Loss Amount of Critical Asset: We then measure the effect of increasing the amount of defender's financial loss or equivalently attacker's financial gain of the first critical asset (i.e., asset with highest loss and gain values). Figure 4 shows such an effect where the attacker (for the same behavioral level) acts more rationally when the critical asset v_1 has a higher financial loss. In particular, Figure 4 shows the QRE probability for attacker approaches PNE choice faster (i.e., at lower λ) when the defender financial loss at the critical asset is higher (red-colored line) compared to the cases of lower loss (blue- and green-colored lines).

Expected Cost of the Defender: The expected costs of defender for simultaneous investments by both defender

¹Note that the five strategies for each player outlined in Table I includes the PNE in which defender $\mathcal{D}$ chooses strategy X_5 and attacker $\mathcal{A}$ chooses strategy Y_5 . Here, we perform the quantal analysis to study the effects of behavioral level on QRE's probabilities and the relationship between QRE and PNE. Thus, we intentionally keep the PNE solution to our set of investment strategies to test whether QRE approaches to PNE or not.

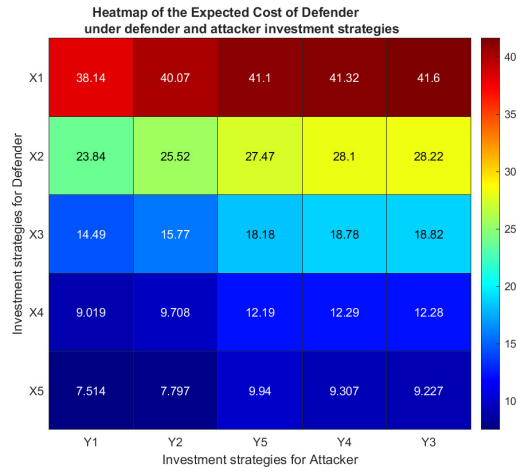


Fig. 5: The heat map of defender's expected cost under defense and attack investment strategies (from Table I).

and attacker has been illustrated through the heat map in Figure 5. It is evident from the heatmap that defender's expected cost is lowest for its best investment strategy (with highest QRE probability) while the attacker chooses any of its investment strategies. Also, the (X_5, Y_1) have the lowest expected cost for the defender $\mathcal{D}$ among all the combinations of the simultaneous attack-defense investment strategies. This also proves the notion that the cost of successful compromise of an asset will be lowest when the defender chooses its best strategy (X_5) while the attacker is choosing the worst (Y_1) .

V. CONCLUSION

We presented a *security investment* model for defenders of modern systems where defender's assets have heterogeneous losses. The proposed security game model captures the existence of behavioral players that have quantal behavior (where they have errors in choosing the best investment strategies). We showed that our proposed game has a quantal response equilibrium (QRE). We then measure the effects of the existence of behavioral defender with quantal errors on the expected cost of that defender. We then illustrated how the QRE can be calculated via simple example. We then evaluated the effects of *behavioral* quantal errors of human decision-makers on system's security through a representative scenario and identified different system parameters that affect the overall system's security for our security game model. There are multiple prospective avenues for future research, including studying heterogeneous types of behavioral players (where attacker and defender have heterogeneous quantal errors) and their resulting impacts, exploring environments in which multiple adversaries can collude to attack the assets, and evaluating our findings empirically via human subject experiments. Building on our current study, these fruitful directions of future research can help in understanding the effects of behavioral decision-makers in real-world systems.

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